

WEEK 5 EXAM PRACTICE

Self-grading sheet, Question 1

Name _____

Date _____

A stab wound cuts through the right half of the spinal cord in the middle of the back, the thoracic part of the cord. Below the injury, what would you expect to find on each side of the body?

- A. Right leg: loss of pain and temperature. Left leg: weakness and loss of position sense.
- B. Right leg: weakness, loss of position sense, and loss of pain and temperature. Left leg: normal.
- C. Right leg: weakness and loss of position sense. Left leg: loss of pain and temperature.
- D. Both legs: weakness and loss of position sense, with pain and temperature normal in both.

The answer is C. The dorsal columns and the corticospinal tract for the right leg run on the right side of the cord at this level, so the right leg loses position sense and strength. Pain and temperature fibers cross within a segment or two of entering, so the right half of the cord carries pain from the left leg, and the left leg loses pain and temperature. This is Brown-Séquard syndrome.

Be honest with yourself. A point counts only if you clearly drew it or said it. If you are not sure you did, it does not count.

Your model: check each point you clearly drew or labeled (6 points)

- ☐ The dorsal column pathway goes up the same side it entered and crosses in the medulla.
- ☐ The spinothalamic pathway synapses in the dorsal horn and crosses within a segment or two of where it entered.
- ☐ The corticospinal pathway crosses at the bottom of the medulla and descends on the same side as the muscles it controls.
- ☐ The cut is on the right half, and the fibers running there are named: dorsal columns from the right leg, corticospinal to the right leg, and spinothalamic fibers carrying pain from the left leg.
- ☐ The losses are mapped to the right legs: right leg weak with no position sense, left leg with no pain or temperature.
- ☐ States that the split pattern puts the lesion in the cord, because above the medulla all three pathways have crossed.

Your teaching: check each point you clearly said (9 points)

- ☐ Read the question and said what it asks.
- ☐ Walked the dorsal column pathway in order.
- ☐ Walked the spinothalamic pathway in order.
- ☐ Walked the corticospinal pathway in order.
- ☐ Showed on the model which fibers the right-sided cut interrupts.
- ☐ A is wrong: it puts every loss on the opposite side, as if each pathway crossed at the other level.
- ☐ B is wrong: it puts every loss on the right, as if pain and temperature did not cross in the cord.
- ☐ D is wrong: losing movement and position sense in both legs needs damage to both halves of the cord.
- ☐ Gave the answer, C, and stated it again at the end.

Model points ____ of 6

Teaching points ____ of 9

Answer right? Yes / No

Your percent ____

Your score: add your model and teaching points, divide by 15, multiply by 75, and add 25 if your answer was right. Example: 5 model points and 7 teaching points is 12 of 15, which is 60; with the right answer, 85 percent.

No video? Grade your model from the photo of your whiteboard and leave the teaching points blank. Your score is then your model points divided by 6, times 75, plus 25 if your answer was right.

WEEK 5 EXAM PRACTICE

Self-grading sheet, Question 2

Name _____

Date _____

A drug blocks phosphodiesterase in the rods of the retina. A flash of light then hits a treated rod. Compared with a normal rod, what happens?

- A. cGMP falls, the cation channels close, the rod hyperpolarizes, and glutamate release falls
- B. cGMP stays high, the cation channels stay open, the rod stays depolarized, and glutamate release stays high
- C. cGMP rises, the cation channels close, the rod hyperpolarizes, and glutamate release rises
- D. cGMP stays high, the cation channels close, the rod hyperpolarizes, and glutamate release falls

The answer is B. Light normally works by having phosphodiesterase break down cGMP. With phosphodiesterase blocked, cGMP stays high, the cGMP-gated channels stay open, Na⁺ keeps coming in, the rod stays depolarized, and glutamate release stays high. The bipolar cell receives no sign that light arrived.

Be honest with yourself. A point counts only if you clearly drew it or said it. If you are not sure you did, it does not count.

Your model: check each point you clearly drew or labeled (6 points)

- ☐ In the dark: cGMP high, channels open, Na⁺ entering, rod depolarized, near minus 40 mV.
- ☐ In the dark the rod releases glutamate continuously.
- ☐ The chain in order: photon, retinal changes shape, rhodopsin active, transducin, phosphodiesterase, cGMP broken down.
- ☐ The block is placed at phosphodiesterase, so cGMP is not broken down and stays high.
- ☐ With cGMP high, the channels stay open, Na⁺ keeps entering, and the rod stays depolarized.
- ☐ Glutamate release stays high, so the bipolar cell receives no signal that light arrived.

Your teaching: check each point you clearly said (9 points)

- ☐ Read the question and said what it asks.
- ☐ Explained the rod in the dark and why it is depolarized.
- ☐ Walked the normal light cascade in order.
- ☐ Explained what phosphodiesterase normally does.
- ☐ Traced the block to cGMP, then the channels, then the membrane potential.
- ☐ A is wrong: it is the normal response to light, which needs a working phosphodiesterase.
- ☐ C is wrong: closing the channels needs cGMP to fall, and a hyperpolarized rod releases less glutamate, not more.
- ☐ D is wrong: with cGMP still high, the channels cannot close, so nothing after that step can happen.
- ☐ Gave the answer, B, and stated it again at the end.

Model points ____ of 6

Teaching points ____ of 9

Answer right? Yes / No

Your percent ____

Your score: add your model and teaching points, divide by 15, multiply by 75, and add 25 if your answer was right. Example: 5 model points and 7 teaching points is 12 of 15, which is 60; with the right answer, 85 percent.

No video? Grade your model from the photo of your whiteboard and leave the teaching points blank. Your score is then your model points divided by 6, times 75, plus 25 if your answer was right.